



**PRINCETON
UNIVERSITY**

Department of Civil and Environmental Engineering

Vibration-based SHM

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Guest lecturer: Esmail Ghorbani, Ph.D., P.Eng.

Overview

- Structural Health Monitoring (SHM)
 - Monitoring the condition, to identify any damage in a structure
 - Evaluation of its remaining safe life span.
- Damage is defined as any changes to:
 - The material or geometric properties
 - The boundary conditions, or
 - The system connectivity
- Damage changes the stiffness, mass, or energy dissipation properties of a system
 - Affects vibration response features
 - Change of the features over time indicate damage (if we are lucky!).

Foundation for vibration-based damage identification methods

- Damage Identification (DI) steps:
 - Detect the damage
 - Localize the damage
 - Quantify the damage

Vibration analysis

- Vibration-based analysis are usually divided into two categories:

Data-based methods

Only use vibration data
Detect and localize damage,
Cannot easily quantify the level of damage.
(like modal analysis methods)

Model-based methods

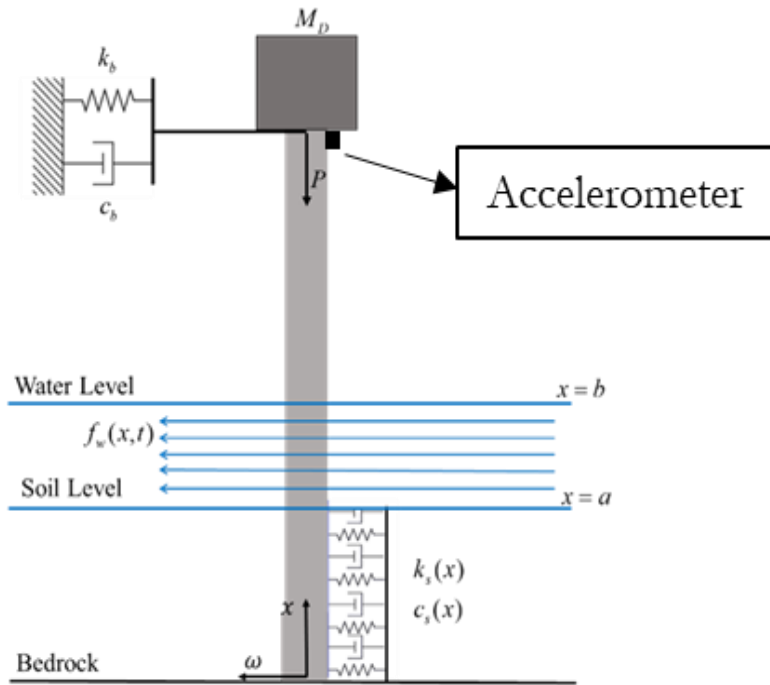
Need a mathematical model.
Able to quantify the level of damage based on the model parameter
Like Kalman filters.

Hybrid methods

Combine model-based and data-based methods

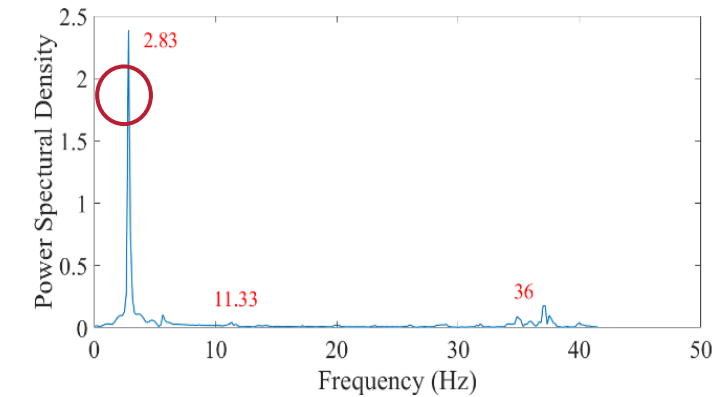
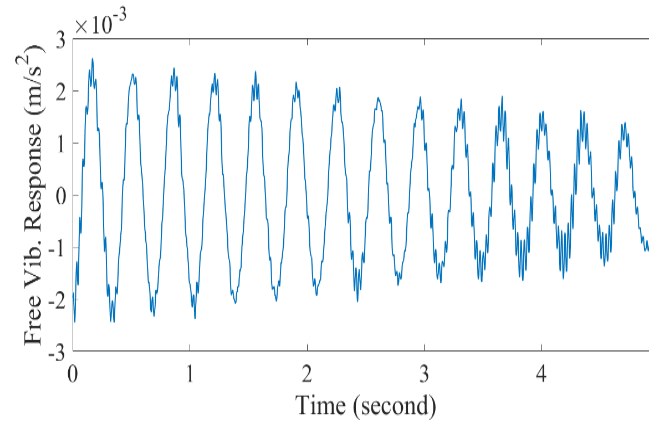
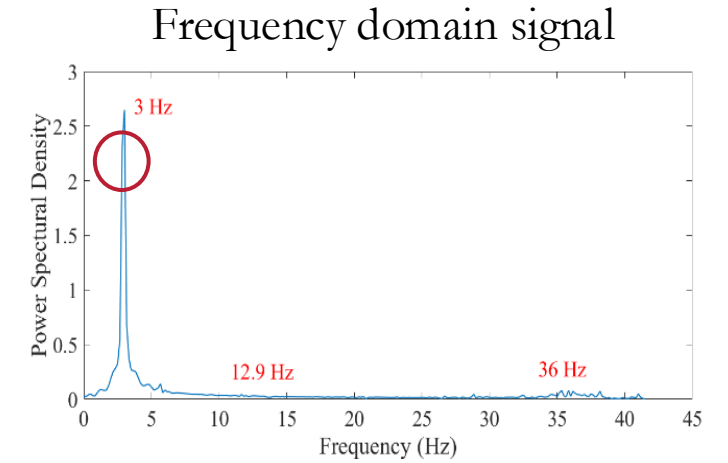
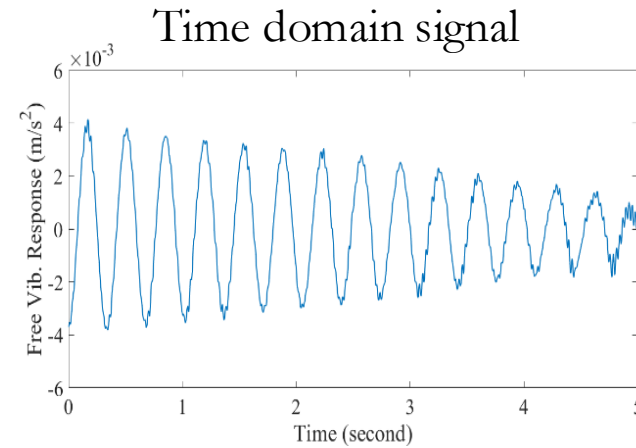
Lets start with an example

I simulated a bridge pier under random excitation, and measured the acceleration of vibration at the tip of pier close to the girder bearing.



Original model

20% change in
stiffness



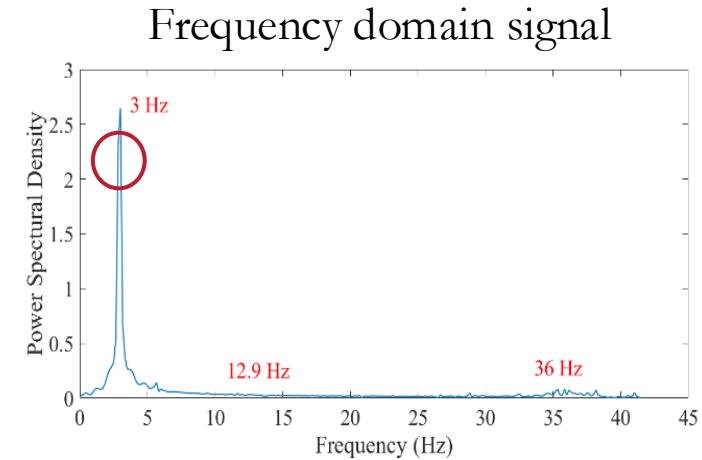
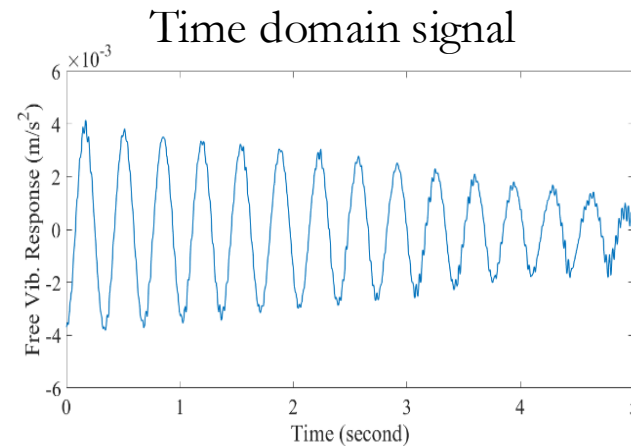
Here, you can ask many questions, which I am going to answer through the lecture!

Q1- How did I developed the simulation model?

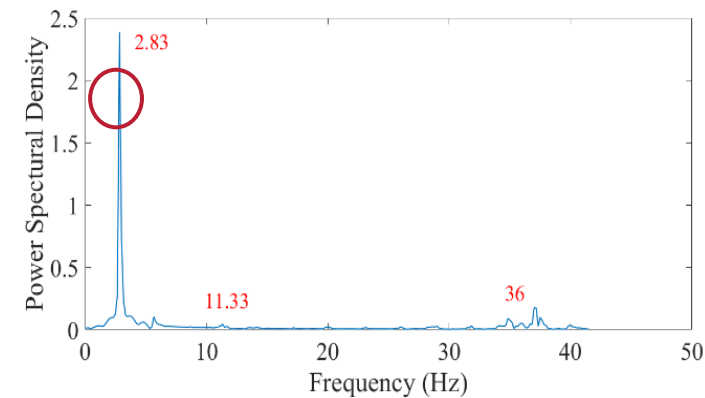
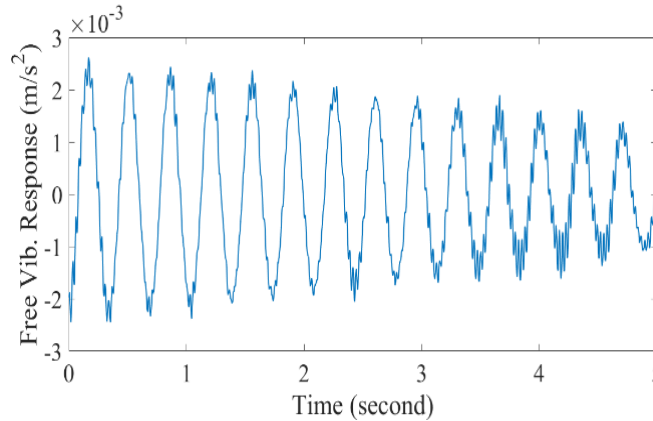
Q2- what are TD and FD?

Q3- Do you have any other question?

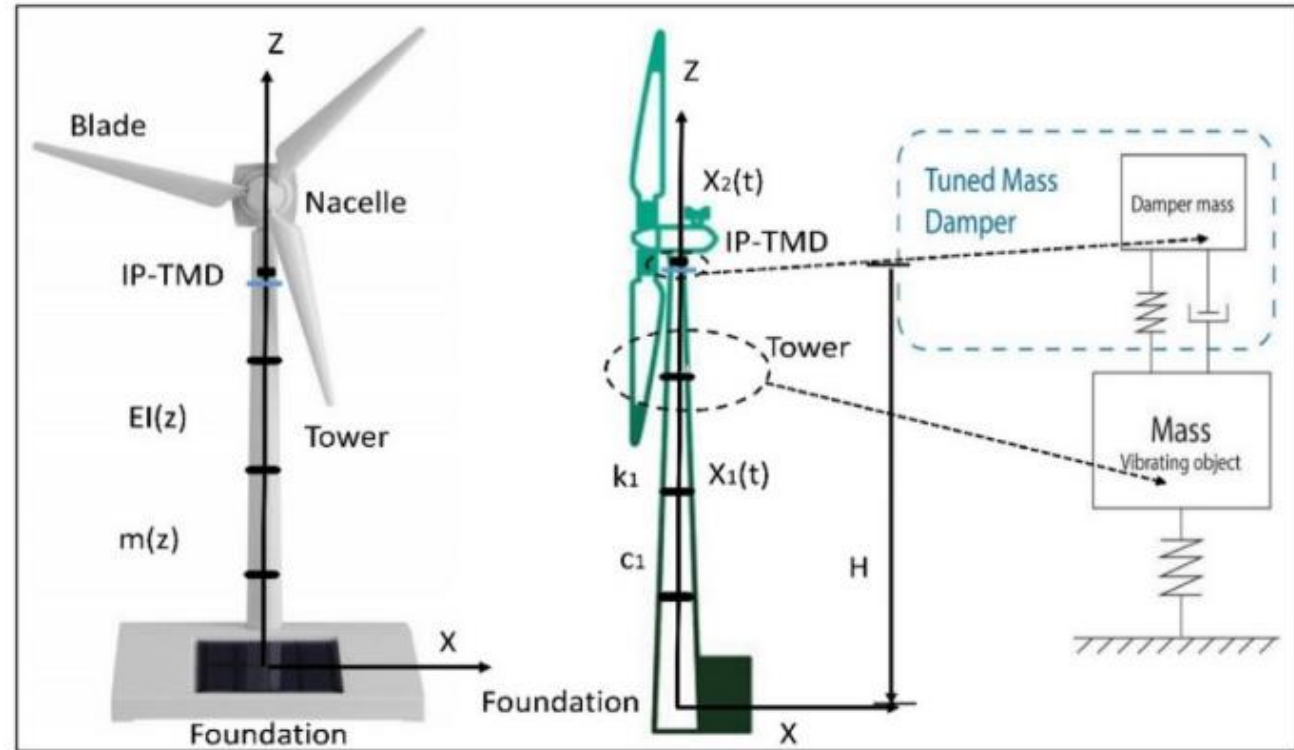
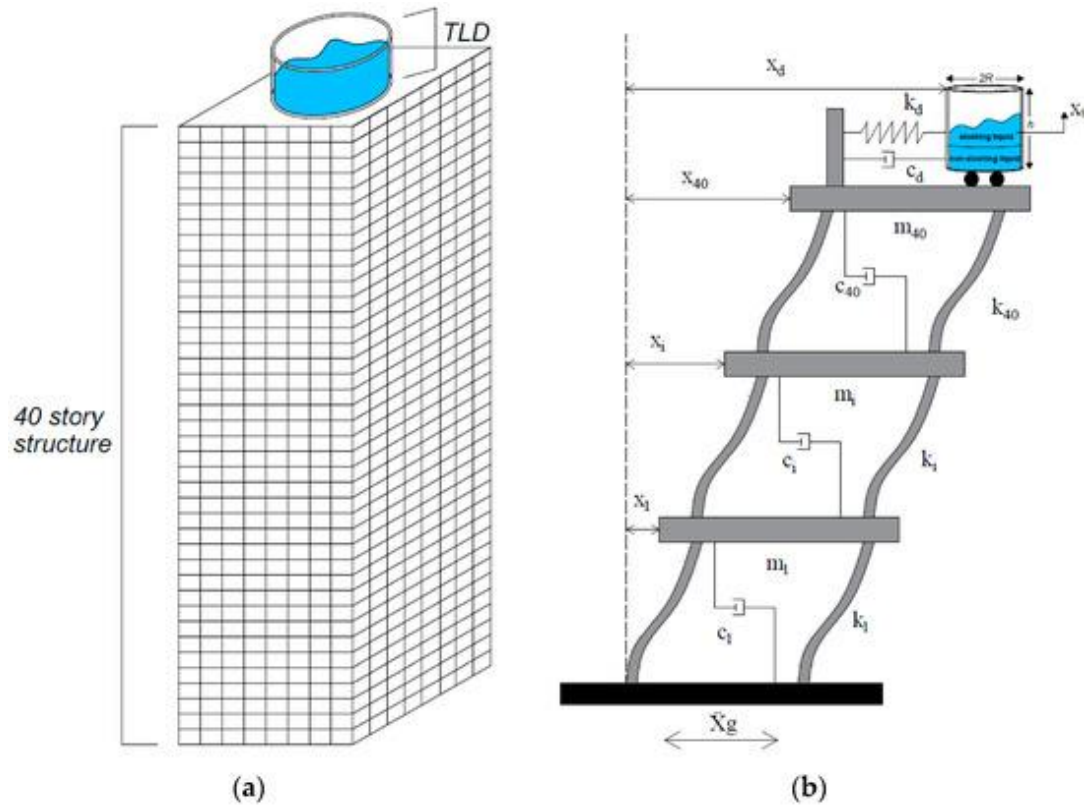
Original model



20% change in
stiffness



More examples



Amer, Y. A., A. T. El-Sayed, and M. M. Agwa. "Controlling wind turbine tower vibration under external force by applying control systems combination." *Scientific Reports* 14.1 (2024): 18597.

Ocak, Ayla, et al. "Optimization of tuned liquid damper including different liquids for lateral displacement control of single and multi-story structures." *Buildings* 12.3 (2022): 377.

Part one- direct problem

Developing governing equation of a one-dof and multi-dof vibrating system

To define a vibrational model

In every dynamical system we have three main physical components:

- The **mass** (m) which resists acceleration according to Newton's second law.

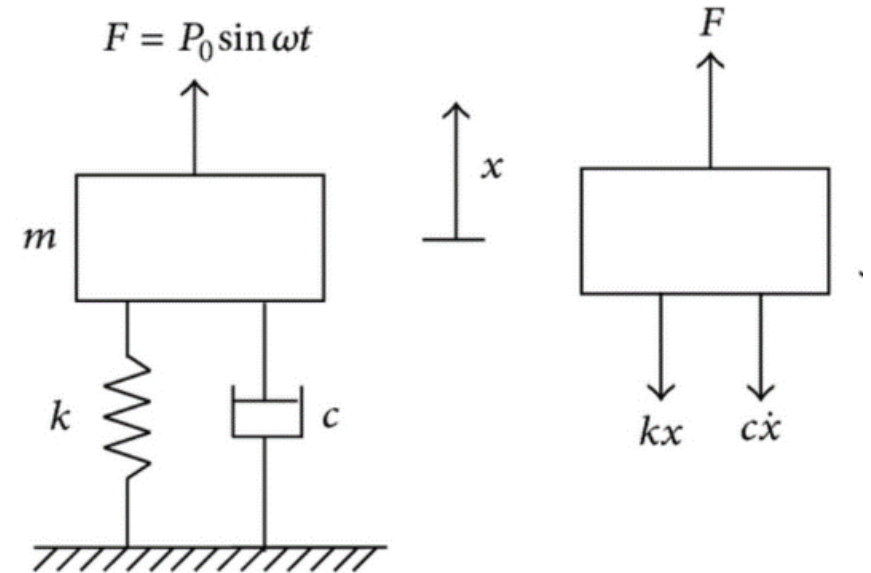
$$F = m\ddot{x}$$

- The **damper** (c) which produces a force proportional to velocity

$$F_d = c\dot{x}$$

- The **spring** (k) which produces a force proportional to displacement (Hooke's law).

$$F_s = kx$$



One degree of freedom system

$$\sum F = m\ddot{x} \Rightarrow F(t) - kx - c\dot{x} = m\ddot{x}$$

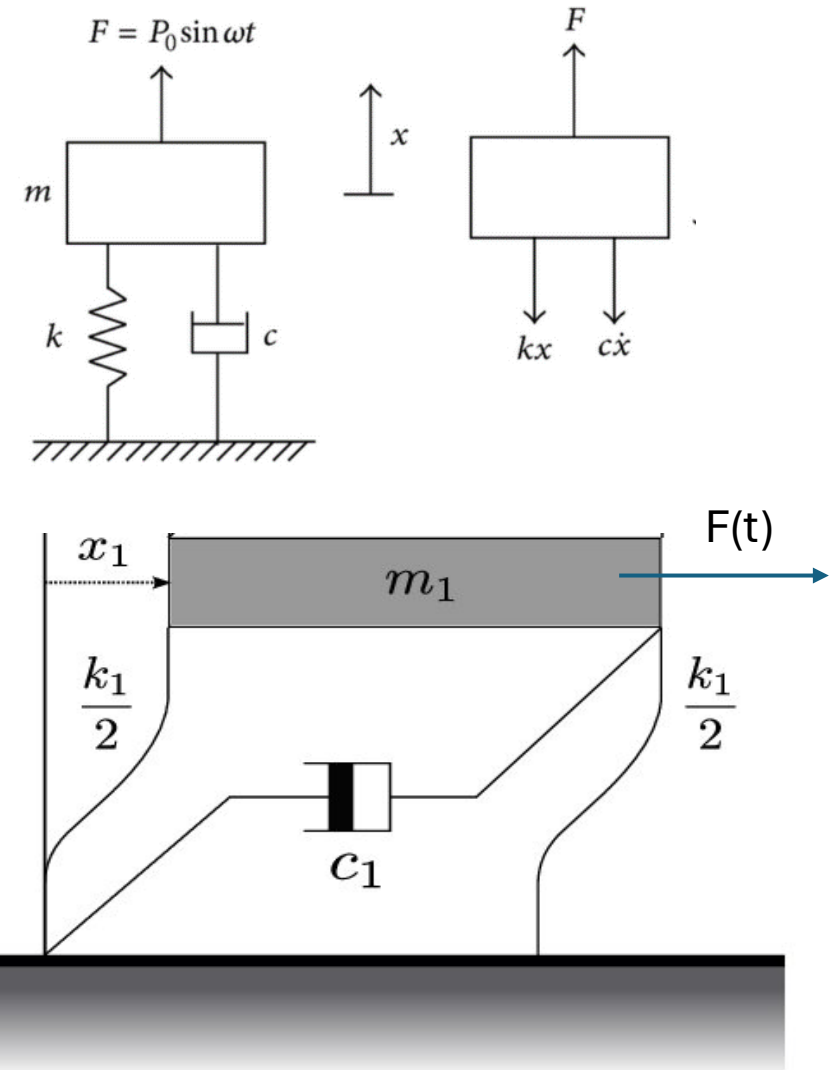
$$m\ddot{x} + c\dot{x} + kx = F(t)$$

$$\Rightarrow \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = \frac{F(t)}{m},$$

$$\text{Define } \omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{2m\omega_n},$$

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F(t)}{m},$$

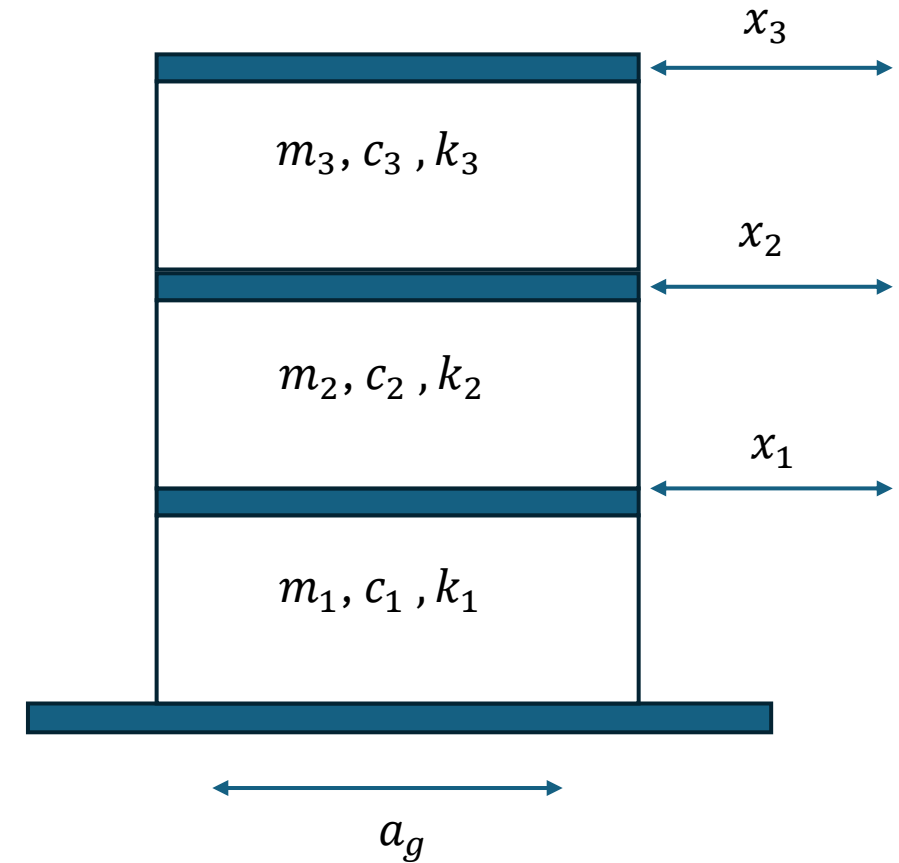
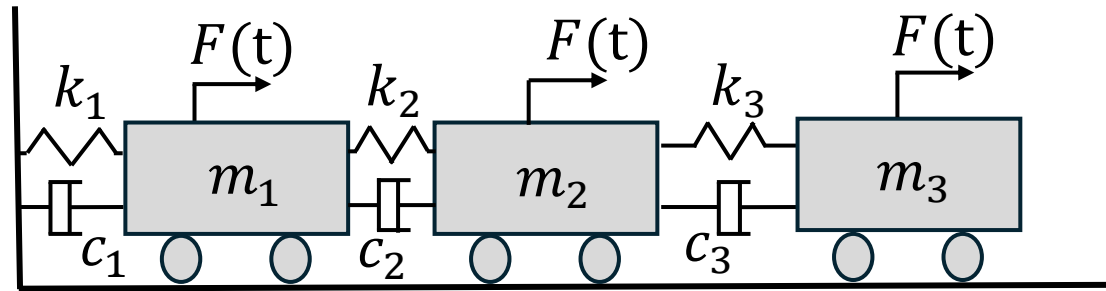
Open the [One_DoF_System.html](#) file



How can we map the modeling to bigger systems?



Lets start with this model.

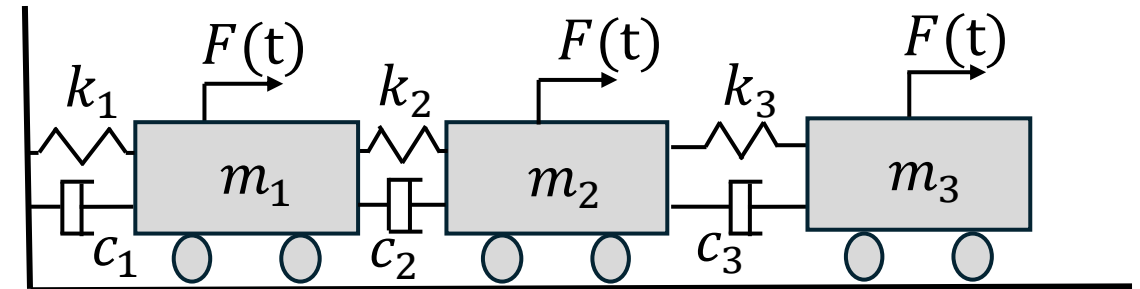


Governing equation for a 3-DoF system

$$m_1\ddot{x}_1 + (c_1 + c_2)\dot{x}_1 - c_2\dot{x}_2 + (k_1 + k_2)x_1 - k_2x_2 = F(t)$$

$$m_2\ddot{x}_2 - c_2\dot{x}_1 + (c_2 + c_3)\dot{x}_2 - c_3\dot{x}_3 - k_2x_1 + (k_2 + k_3)x_2 - k_3x_3 = F(t)$$

$$m_3\ddot{x}_3 - c_3\dot{x}_2 + c_3\dot{x}_3 - k_3x_2 + k_3x_3 = F(t)$$



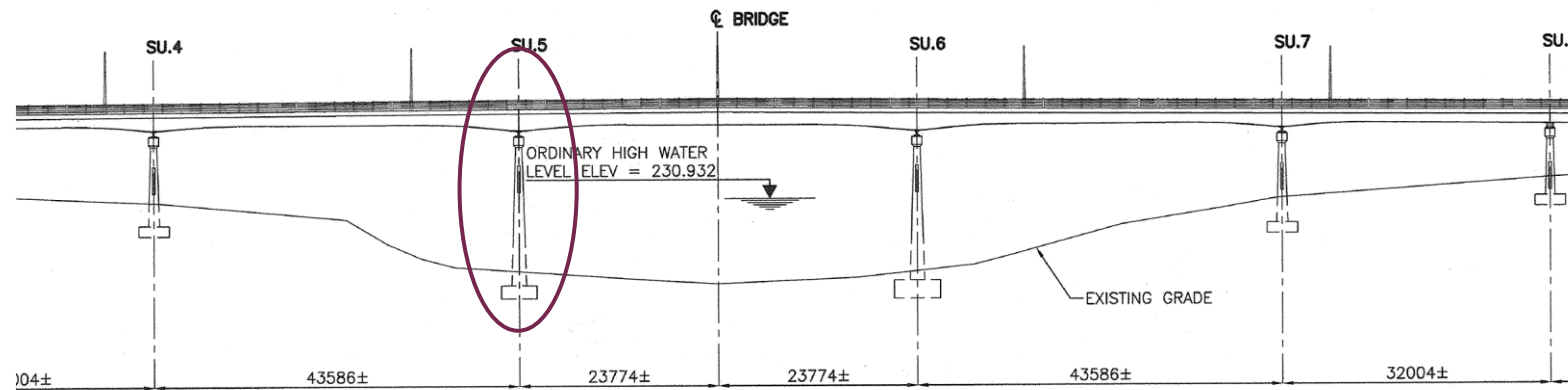
- Can we write this in matrix form?
- How can we solve this system of equation?
- Can we write this system of equation in standard form similar to one DoF system?

Open the `Three_DoF_System.html` file

Part two-inverse problem

In this part, we assume that we measure the vibration, how can we extract some features like natural frequency and damping value from the data.

Vibration response of a 5-span bridge



BRIDGE SOUTH ELEVATION



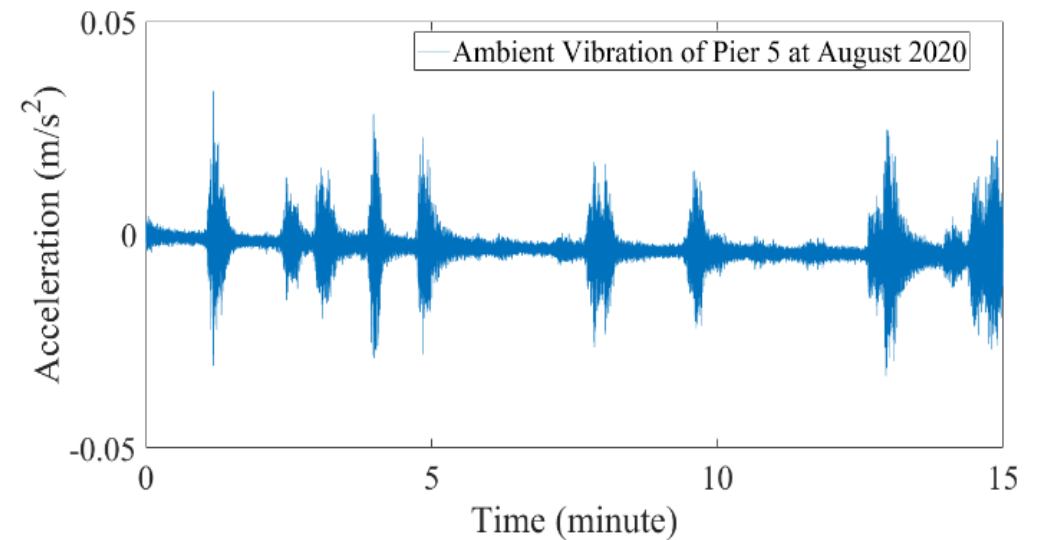
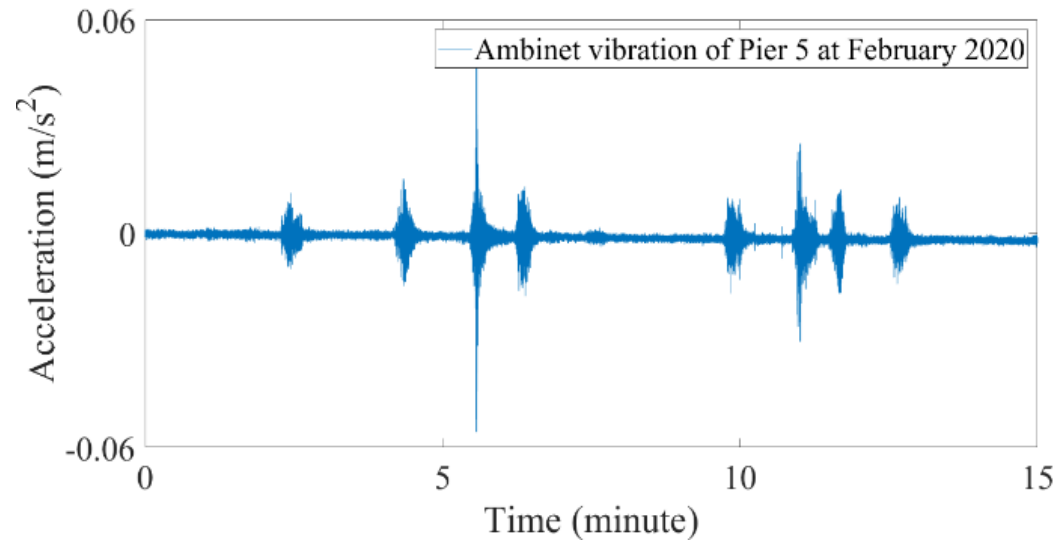
Morris bridge, 5-span, 300 meter long bridge located in Morris city, Manitoba, Canada, 2019-2021

Vibration response of a 5-span bridge

As you see here, the vibration response is very complicated and its shape is not close to what we saw so far.

There is one big issue here, the vibration response amplitude is depended to the car size crossing the bridge.

Raw vibration data



What is correlation function?

The **correlation function** measures the similarity between two signals as a function of a time shift.

- For two signals $x(t)$ and $y(t)$, the **cross-correlation function** is defined as

$$R_{xy}(\tau) = E[x(t) y(t + \tau)]$$

- $E[\cdot]$ =expectation (statistical average)
- τ = time lag
- $R_{xy}(\tau)$ =correlation between the signals at delay τ
- For discrete signals, the code estimates:

Open the corr_function.html file

$$R_{x\dot{x}}(\tau) \approx \frac{1}{N} \sum_{n=0}^{N-1} x(n) \dot{x}(n + \tau)$$

Random Decrement

The **Random Decrement (RD)** technique is a signal processing method used to extract the **free vibration response** of a structure (like a bridge, building, or machine) excited by unknown, random ambient forces (like wind, traffic, or waves).

For a linear, time invariant system, with random and stationary excitation:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = f(t)$$

To derive the RD, we multiply the equation of motion by $x(t + \tau)$:

$$m\ddot{x}(t)x(t + \tau) + c\dot{x}(t)x(t + \tau) + kx(t)x(t + \tau) = f(t)x(t + \tau)$$

Taking the expectation $E[\cdot]$:

$$mE[\ddot{x}(t)x(t + \tau)] + cE[\dot{x}(t)x(t + \tau)] + kE[x(t)x(t + \tau)] = E[f(t)x(t + \tau)]$$

Random Decrement

These expectations become correlation functions:

$$E[x(t)x(t + \tau)] = R_{xx}(\tau), \quad E[\dot{x}(t)x(t + \tau)] = \frac{d}{d\tau} R_{xx}(\tau), \quad E[\ddot{x}(t)x(t + \tau)] = \frac{d^2}{d\tau^2} R_{xx}(\tau)$$

The correlation function satisfies the **same homogeneous equation as free vibration**

$$m \frac{d^2 R_{xx}(\tau)}{d\tau^2} + c \frac{dR_{xx}(\tau)}{d\tau} + kR_{xx}(\tau) = 0$$

The **correlation function behaves like the free decay response of the system.**

How to implement Random Decrement

1- Lets define a **trigger condition** $x(t_i) = x_0$

2- The points that the signal satisfy the trigger:
 $t_1, t_2, t_3, t_4, t_5, t_6$.

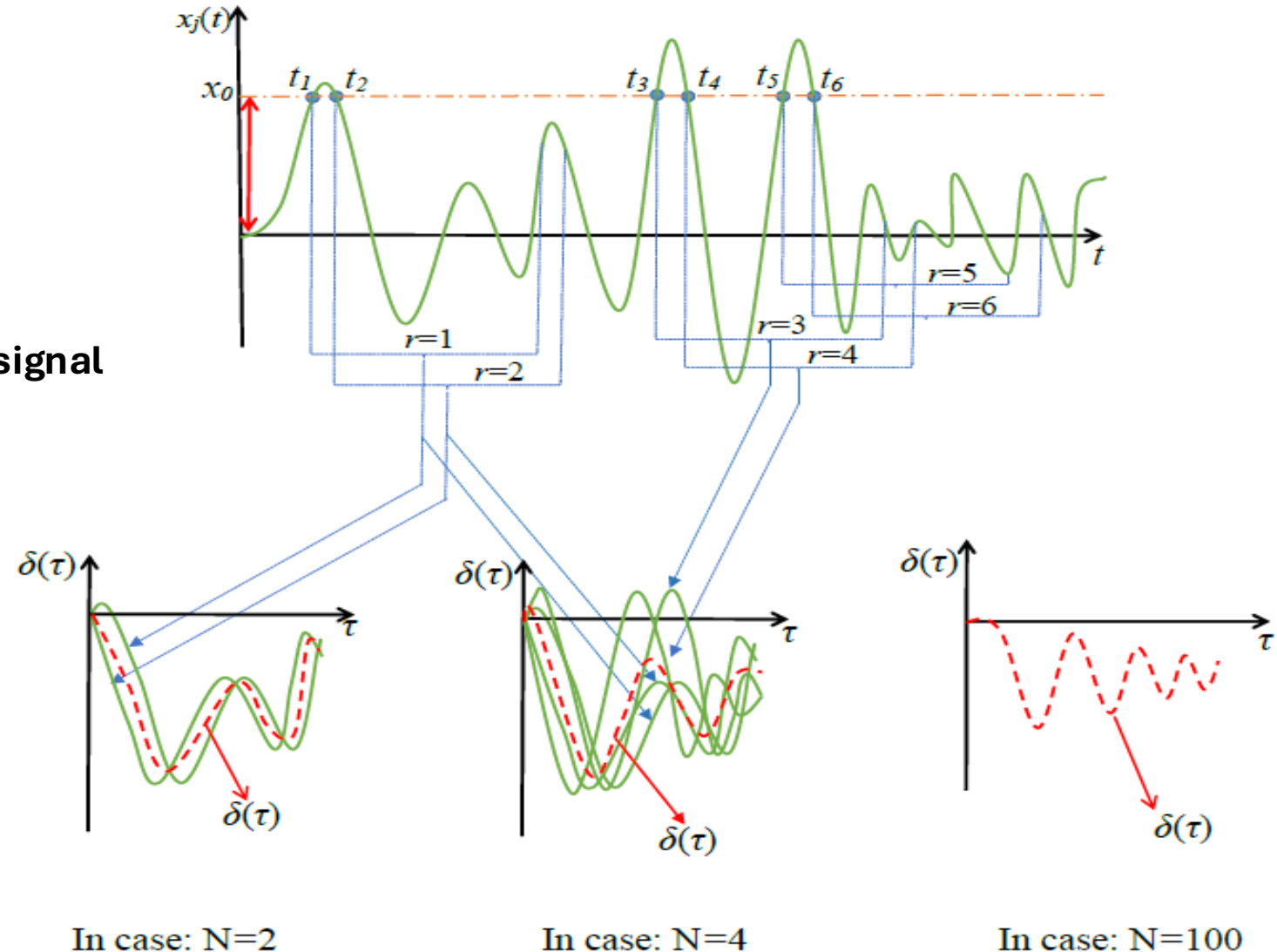
3- For each trigger time we take a **segment of the signal**
 $x(t_i + \tau)$

The random decrement average is

$$\delta(\tau) = \frac{1}{N} \sum_{i=1}^N x(t_i + \tau)$$

This is exactly what the bottom plots show:

$N = 2 \rightarrow$ noisy
 $N = 4 \rightarrow$ cleaner
 $N = 100 \rightarrow$ converges



Relation between correlation and averaging segments

At each trigger: $x(t_i) = x_0$

So the correlation definition becomes: $R_{xx}(\tau) = E[x(t)x(t + \tau)]$

Conditioning on $x(t) = x_0$: $R_{xx}(\tau) = x_0 E[x(t + \tau) | x(t) = x_0]$

But, $E[x(t + \tau) | x(t) = x_0]$ is exactly what averaging the segments.

Thus $\delta(\tau) \propto R_{xx}(\tau)$

More precisely: $\delta(\tau) = \frac{1}{x_0} R_{xx}(\tau)$

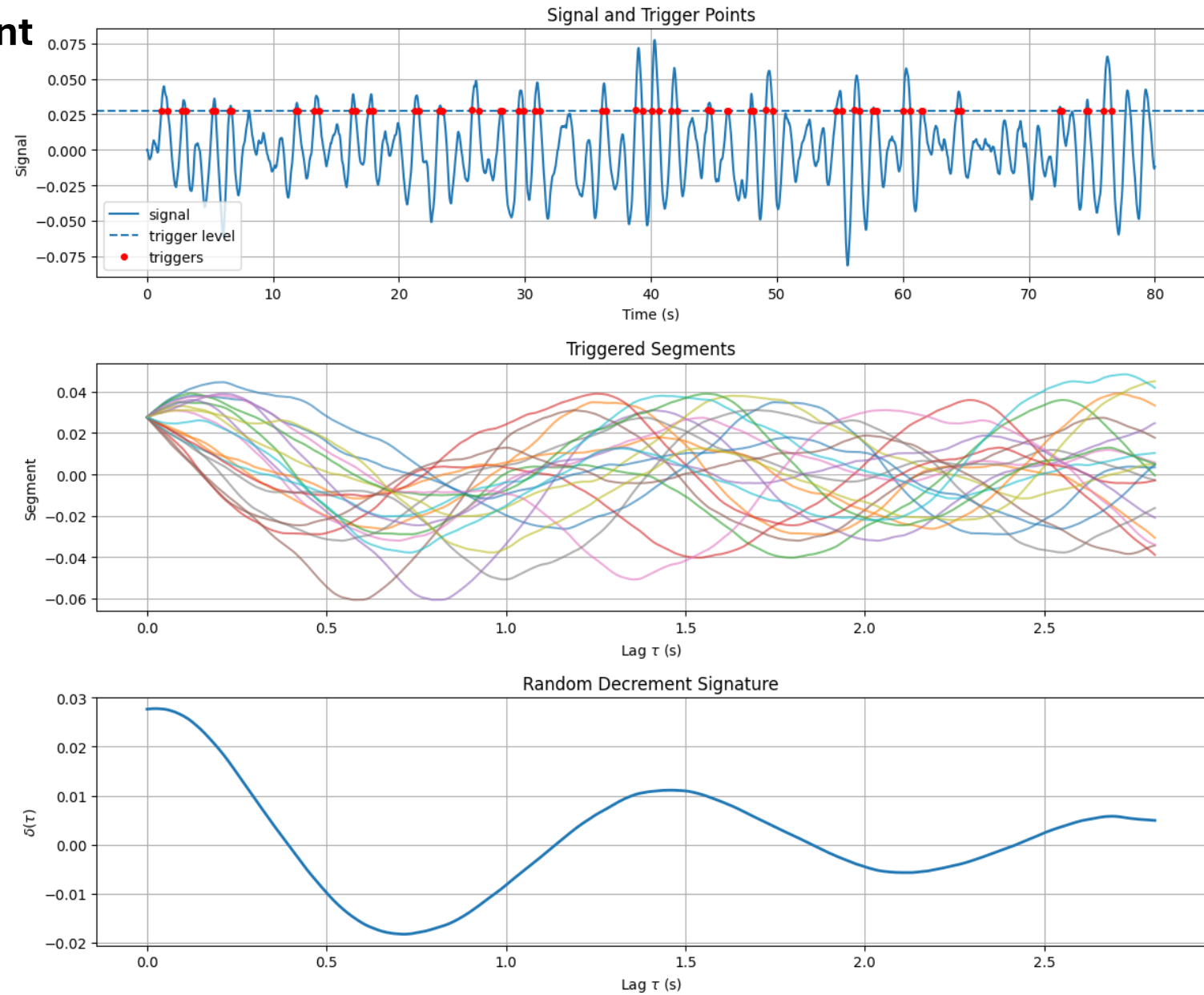
Self-study

Simulation results for Random Decrement

Common RD Triggering Methods

- Upward level crossing
- Downward level crossing
- Bidirectional level crossing
- Absolute amplitude trigger $|x| = L$
- Peak (local maximum) trigger
- Multi-condition trigger (e.g., $x = L$ and $\dot{x} > 0$)

Open the `RD_algorithm.html` file



How to use RD signatures?

For a 1 DoF system

Natural Frequency

Estimate the **damped period** from peak spacing:

$$1. \quad T_d = \tau_{i+1} - \tau_i,$$

$$2. \quad f_d = \frac{1}{T_d}$$

$$3. \quad \omega_d = 2\pi f_d$$

For lightly damped systems: $\omega_n \approx \omega_d$

Damping Ratio

Logarithmic Decrement

Using peak amplitudes of the RD signal:

$$\delta = \ln \left(\frac{x_1}{x_2} \right)$$

Using multiple cycles:

$$\delta = \frac{1}{n} \ln \left(\frac{x_1}{x_{n+1}} \right)$$

Damping Ratio

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

For small damping:

$$\zeta \approx \frac{\delta}{2\pi}$$

Simulation results for Random Decrement

Damped period from peaks

$$T_d \approx 1.26 - 0.63 \approx 0.63 \text{ s}$$

Natural frequency

$$f_n \approx f_d = \frac{1}{T_d} = \frac{1}{0.63} \approx 1.6 \text{ Hz}$$
$$\omega_n \approx 2\pi f_n \approx 10.1 \text{ rad/s}$$

Logarithmic decrement

Using two peaks

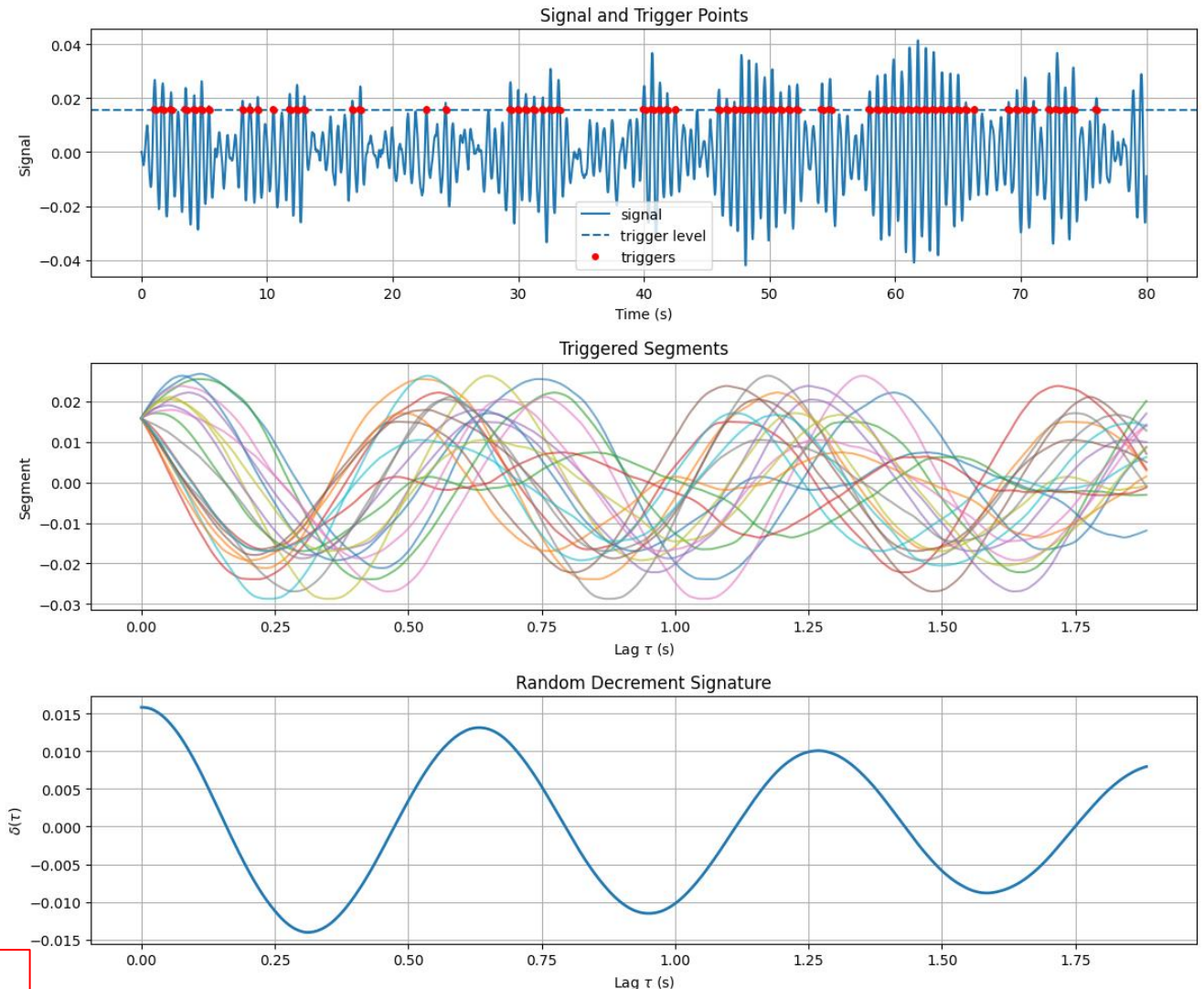
$$x_1 \approx 0.016, x_2 \approx 0.013$$
$$\delta = \ln\left(\frac{0.016}{0.013}\right) \approx 0.21$$

Damping ratio

$$\zeta \approx \frac{\delta}{2\pi} = \frac{0.21}{6.283} \approx 0.033$$

Can you modify the RD code for the 3DoF system?

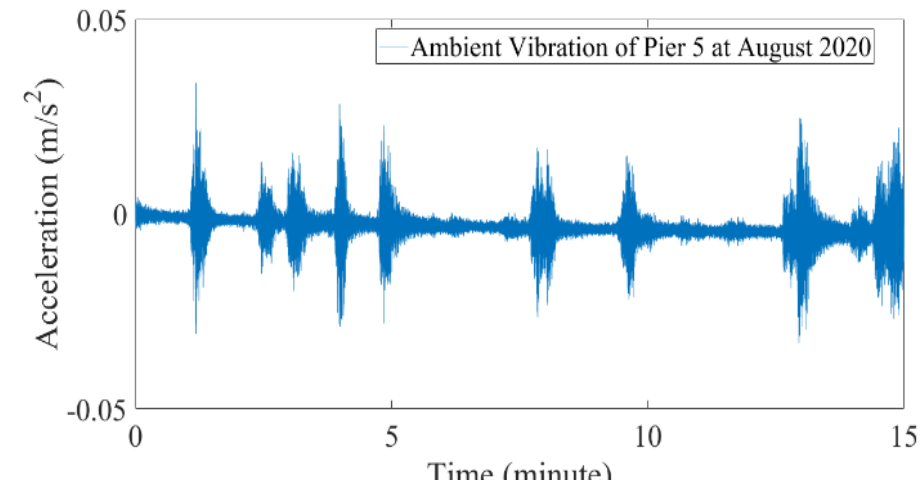
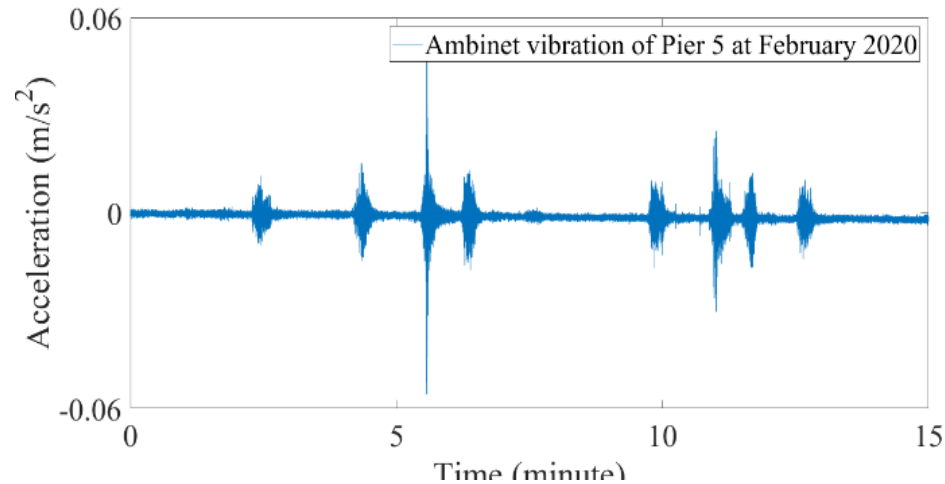
$m = 1.0 \text{ kg}, c = 0.8 \text{ N.s/m}, k = 100 \text{ N/m}$



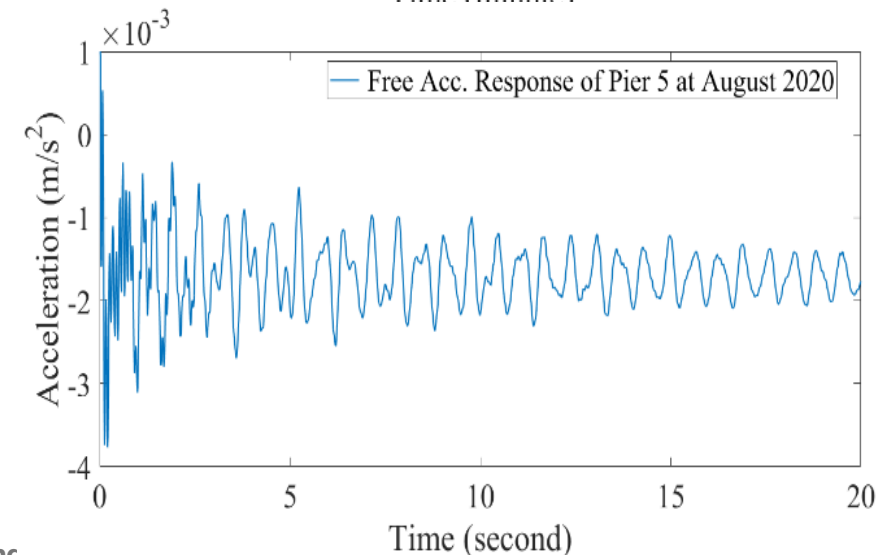
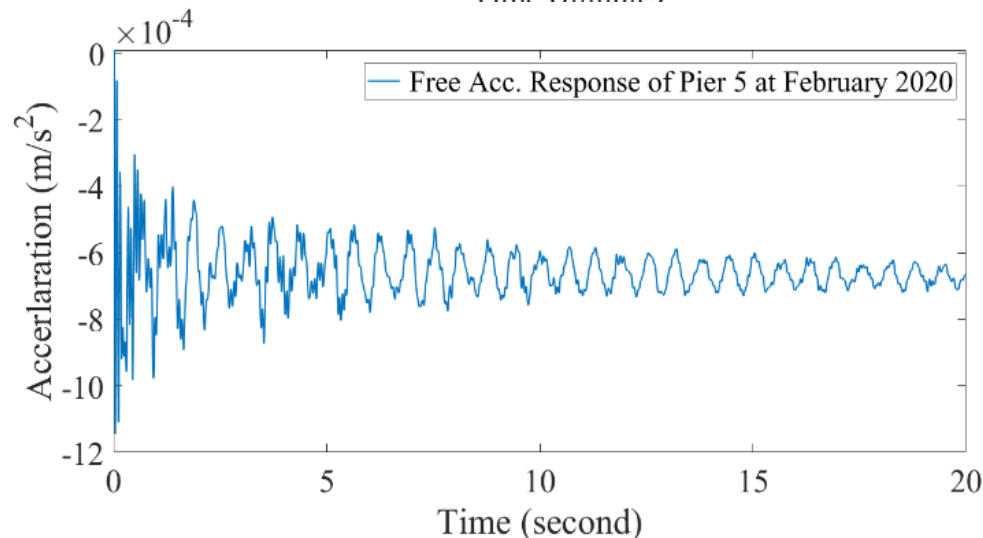
RD on the full-scale bridge data

There are many frequency involved in the response. It is not a simple solo-frequency free vibration response.

Raw vibration data



Free vibration response



Ibrahim time domain method

For 1 DoF system, we can use logarithmic decrement, **what about** multi-dof system, or the bridge which we just saw its free vibration response? **How can we extract the modal parameters for these systems?**

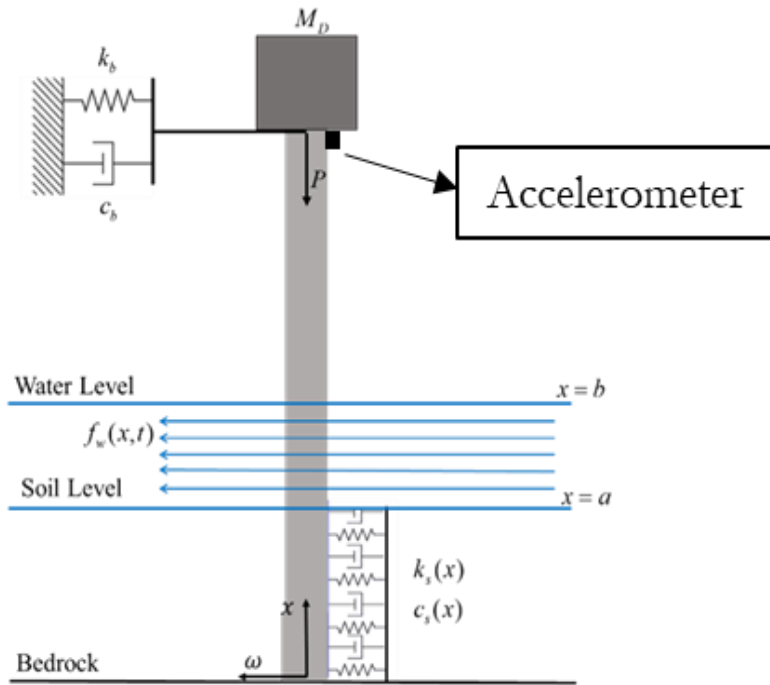
To answer to this question, there is another technique names Ibrahim time domain methods that use the free vibration response of any dynamical system to extract the modal parameters.

The math behind the algorithm and one example is implemented in the following file.

Open the ITD_algorithm.html file

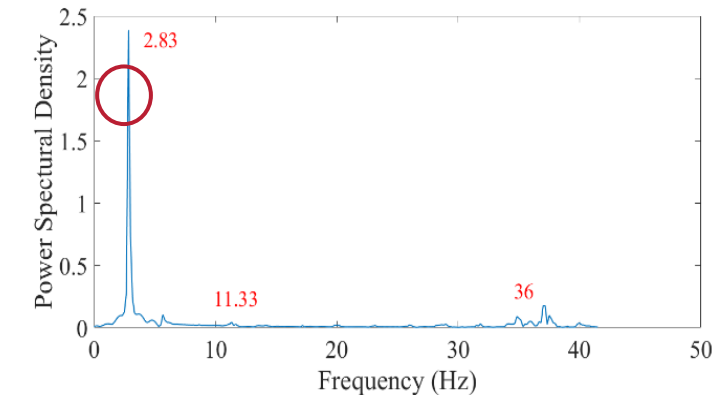
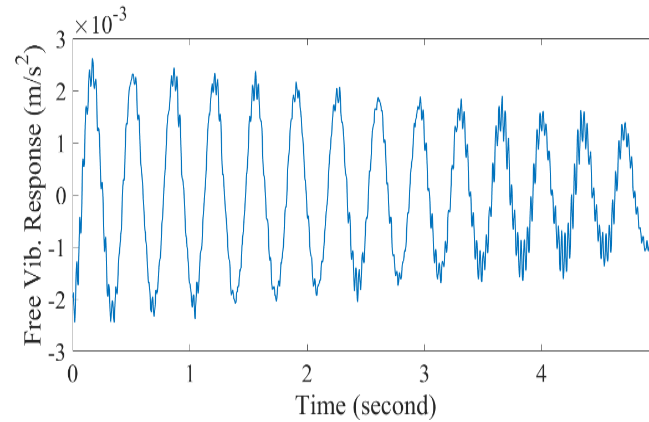
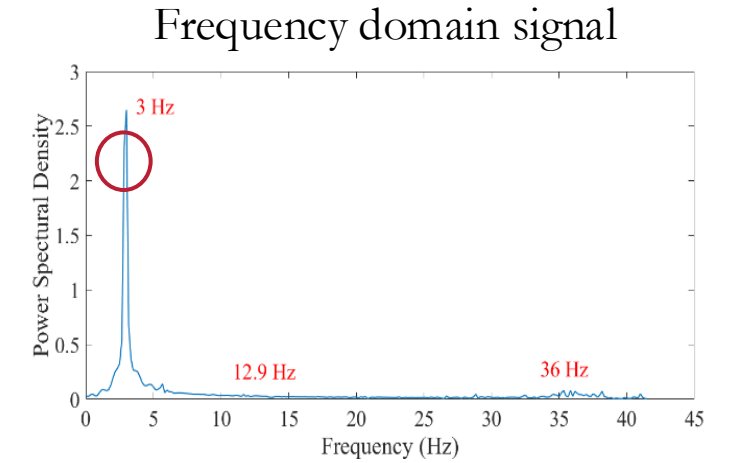
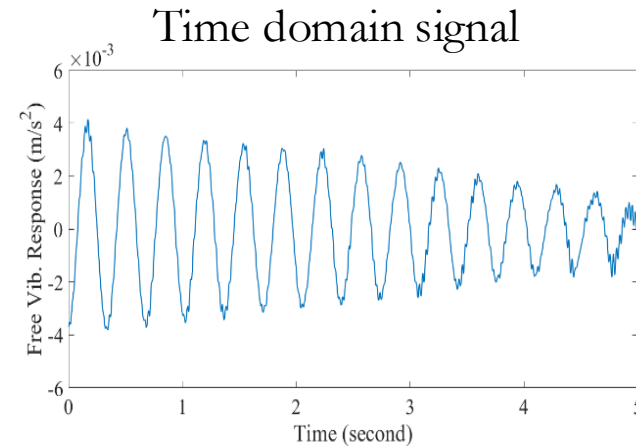
Recall

So far, we can recognize the free vibration response (in time domain). However, what is the FD and how can we travel ☺ there from TD?



Original model

20% change in
stiffness



Frequency domain

In the **frequency domain**, instead of showing a signal as a function of **time**, we represent it by its **frequency components**.

These frequency components are related to the **physical characteristics of the system**, such as **mass, stiffness, and damping**.

If these characteristics change, the **frequency content also changes**, which means something in the system is changing.

In simple terms, the **peaks in the frequency spectrum** often correspond to the **natural frequencies** of the system for a given measured signal.

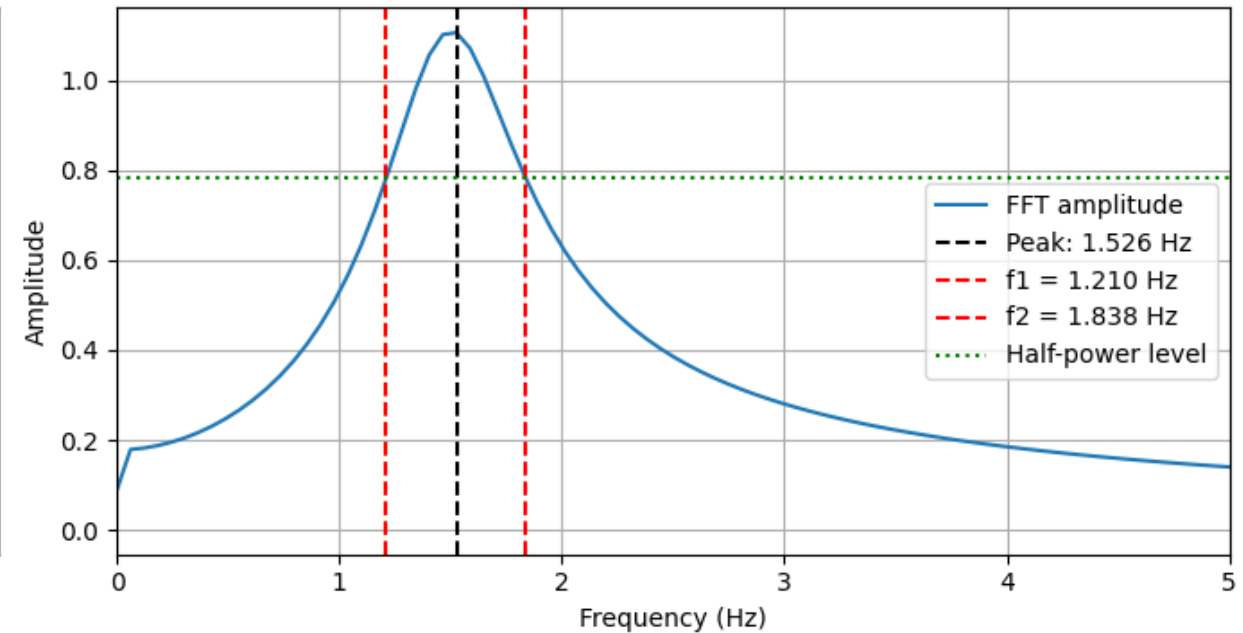
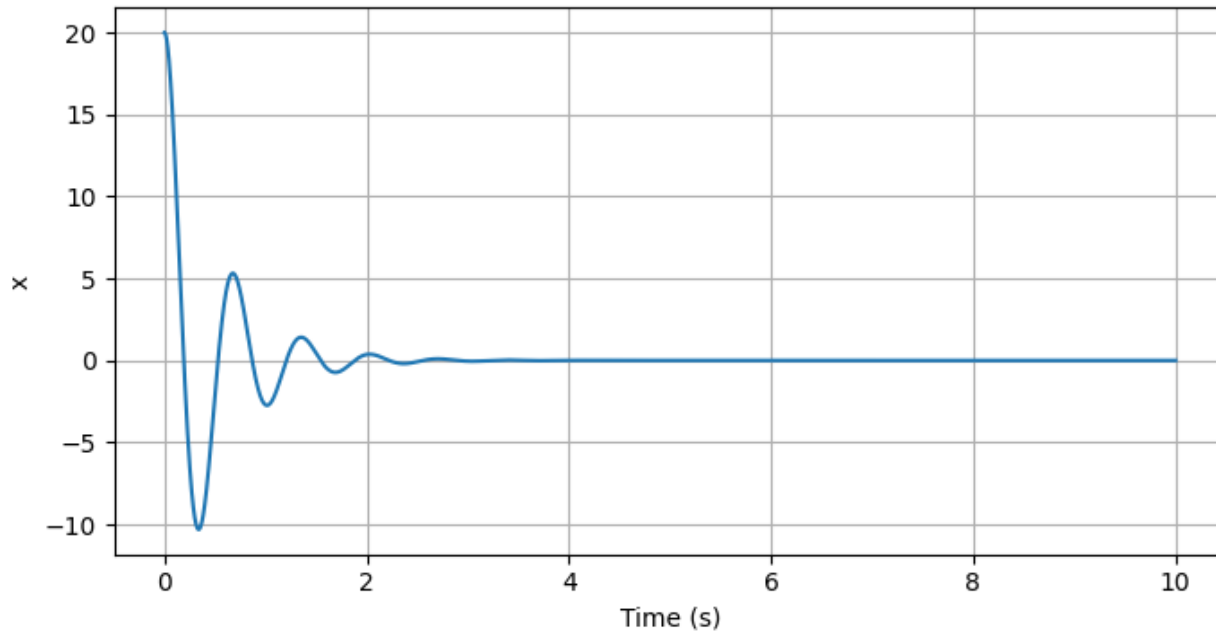
For example, if a **crack initiates**, the **stiffness decreases**. Since the natural frequency depends on stiffness,

$$\omega_n = \sqrt{\frac{k}{m}}$$

a decrease in stiffness causes the **natural frequency to decrease**.

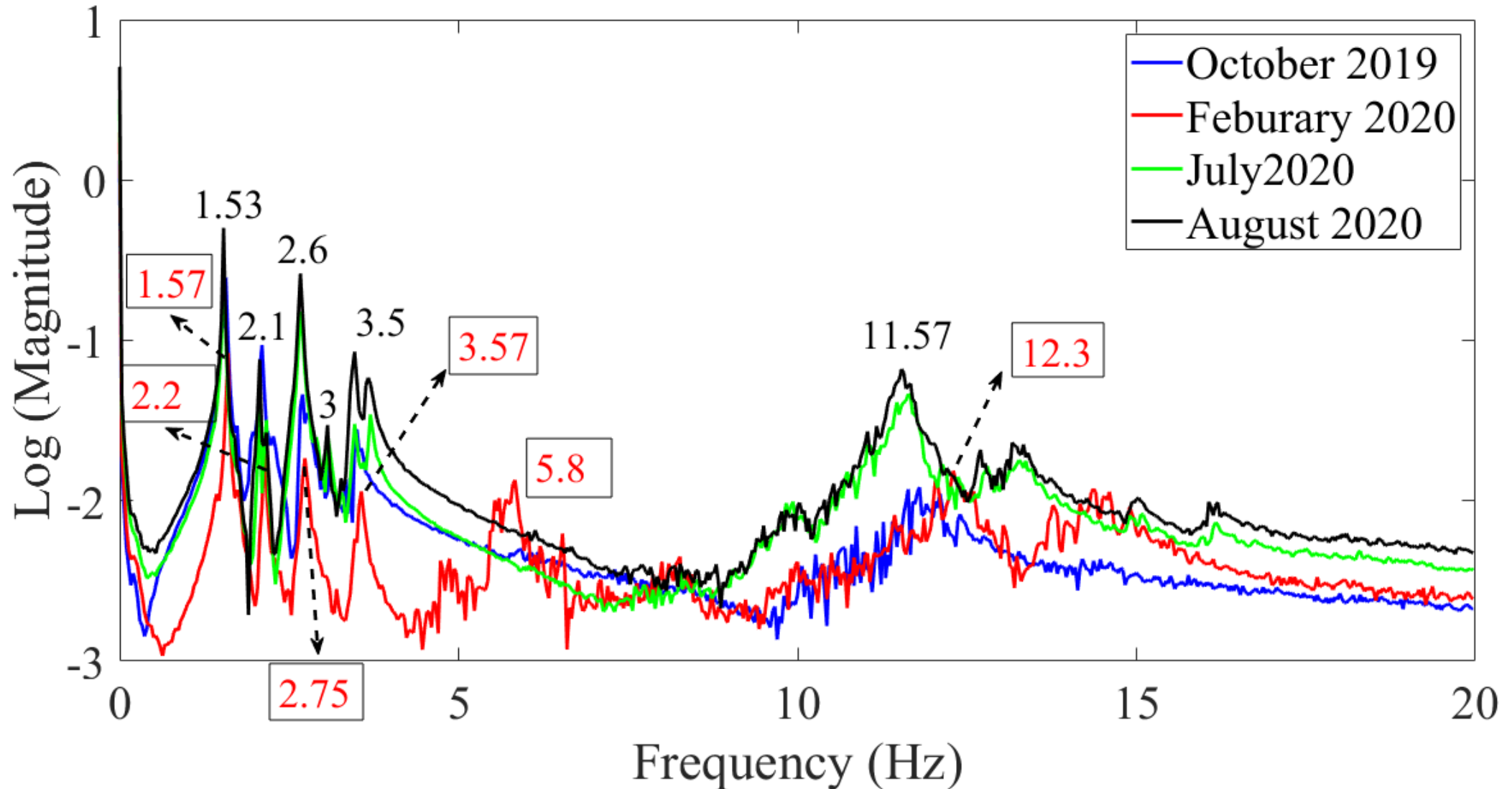
Therefore, by looking only at the **frequency domain**, we can detect that **something is changing in the system**.

Frequency domain



Open the `FD_algorithm.html` file

FD response of the full-scale bridge



RD on the full-scale bridge data



Lesson learnt

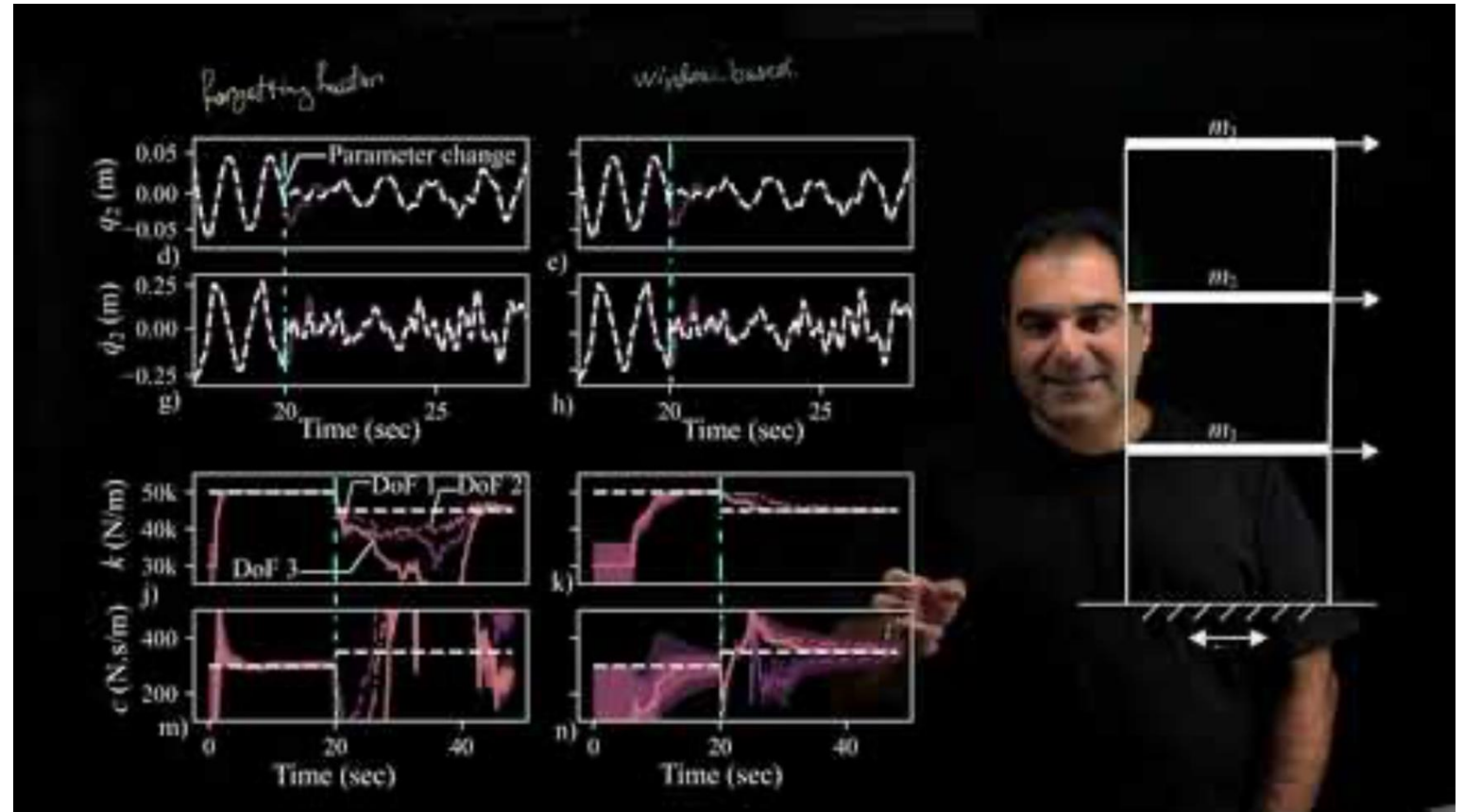
- When we compare the FFT spectra over different months, some peaks shift slightly in frequency.
- A **shift to lower frequencies** may indicate a **reduction in system stiffness**, which could be caused by scour around the pier. However, we also observe that in winter (February) some frequencies appear higher than in other months. This may be explained by the **presence of thick ice accumulated** around the pier, as shown in the pictures.
- The ice effectively adds stiffness and confinement to the structure, which can increase the natural frequencies, causing the peaks to shift slightly to the right in the frequency spectrum.
- Therefore, frequency-domain analysis allows us to **detect that something in the system is changing**.
- However, **FFT alone cannot quantify how much scour or damage has occurred**, so additional analysis is required to estimate the severity of the change or to distinguish environmental effects from actual structural changes.
- This example shows an important point in structural health monitoring:
- changes in frequency do not always indicate damage — they may also be caused by environmental conditions.

Self-study: Model-based SHM

- If you are interested in model-based structural health monitoring, a very good technique is Kalman filtering.
- I have prepared a full course explaining the Kalman filter from scratch to advance structural health monitoring for damage quantification in civil and mechanical systems.
- Please go through and I would be more than happy to talk about, if anyone is interested.



<https://lnkd.in/ejFSVnGt>



Any Question?

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